William Marella

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EDUCATION

University of Cambridge November 2024
M. Phil. Population Health Sciences, Specialization in Health Data Science
Relevant Coursework: Advanced Biostatistics, Bayesian Statistics, Machine Learning, Applied Genomics, Causal Inference, Epidemiology

Brown University May 2023
B.A. Biology, GPA: 3.96

SKILLS

Programming: R, Python, TypeScript, SQL, PostgreSQL, Bash, Git
Go-To Tools: tidyverse, lme4, NumPy/SciPy, scikit-learn, optuna, Bun, React, pgvector
Data Science & ML: Machine Learning, Causal Inference, Frequentist and Bayesian Modeling, AI Agents

EXPERIENCE

Senior Data Scientist New York, New York
Columbia University, Belsky Lab *September 2024 - Present*

- Co-led design and execution of causal inference and ML strategy for an awarded **$10M ARPA-H grant**. Architect and lead developer for statistical R packages used internally and externally by collaborators at **Novo Nordisk and Lilly**
- Served as analytical lead for the GeroScience Computational Core, partnering with faculty across Columbia to design and execute experiments in biological aging and longevity
- Mentored master's students in biostatistics and data science, guiding end-to-end research projects

Visiting Researcher Cambridge, UK
University of Cambridge, Inouye Lab *June 2024 - Present*

- Applied unsupervised Bayesian modeling and scalable inference to identify latent disease groupings from comorbidity data
- Derived mean-field variational inference and partially collapsed Gibbs sampling algorithms. Benchmarked accuracy and time complexity on simulated data designed to mirror real UK Biobank data
- Implemented both methods from scratch in Python (NumPy/SciPy) with convergence diagnostics and multi-threaded MCMC chain coordination. Packaged as a Python library (*GitHub*)

RESEARCH PROJECTS

FraminghamPACE | *R*

- Developed a hierarchical model of the pace of aging and ML training approach to create a state-of-the-art epigenetic aging clock with improved predictive performance, technical reliability, and responsiveness to intervention
- Selected for oral presentation at the Biomarkers of Aging Consortium 2025, Biomarker Network Conference 2026, and The Advances in Social Genomics Conference 2026
- **Patented and licensed algorithm** to TruDiagnostic (**co-lead inventor**); manuscript in preparation

UK Biobank BioScore | *R*

- Collaborated with the founder of the HOOKE Longevity Clinic to construct a novel composite aging score in the UK Biobank (500,000+ participants), independently designing and implementing the measure
- Validated BioScore across retrospective, cross-sectional, and prospective outcomes including time-to-event analyses. Outperformed PhenoAge, the current field standard. Manuscript in preparation

SOFTWARE PROJECTS

QueryGaP | *Python, PostgreSQL, JavaScript* | *Deployed*

- Scraped and indexed all dbGaP phenotype data documentation into PostgreSQL with hybrid semantic and keyword search (pgvector, tsvector) to drastically speed up navigation of many thousands of phenotype variables
- Built an AI agent to query the database and return natural language answers about study documentation, **cutting search time by over 90%**
- Used daily at Belsky lab; institution-wide adoption at Columbia under evaluation

OpenCode Protect | *TypeScript, Node.js, Bun* | *GitHub*

- Existing AI coding agents are granted full filesystem access, with prompt-level protections that are trivially bypassed by prompt injection attacks
- Forked OpenCode to implement kernel-level isolation of AI agent commands, running them as a restricted Unix user to protect sensitive credentials and files

CostcoGuessr | *TypeScript, Next.js, Bun, PostgreSQL* | *Deployed*

- Scraped 10,000+ Costco products across 11 countries, stored in PostgreSQL with images on Vercel Blob
- Built a full-stack daily price-guessing game in TypeScript/Bun, deployed via Railway and Vercel, reaching **200 daily users**

Open-Source Contributions: *TypeScript, React, Bun, Node* | *PR#1*, *PR#2*

- Merged navigation and UI features and fixes into OpenCode and PostHog codebases